第七单元 弯曲应力

纯弯曲时的正应力

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内容提要

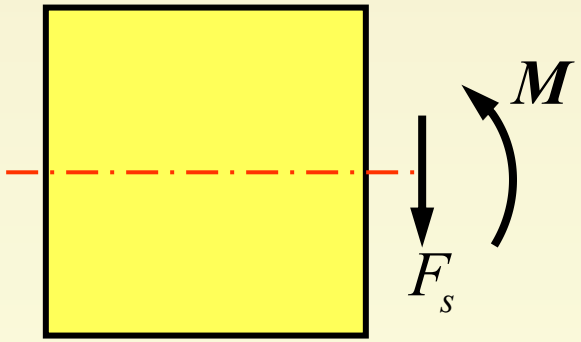
01 弯曲时横截面上的应力

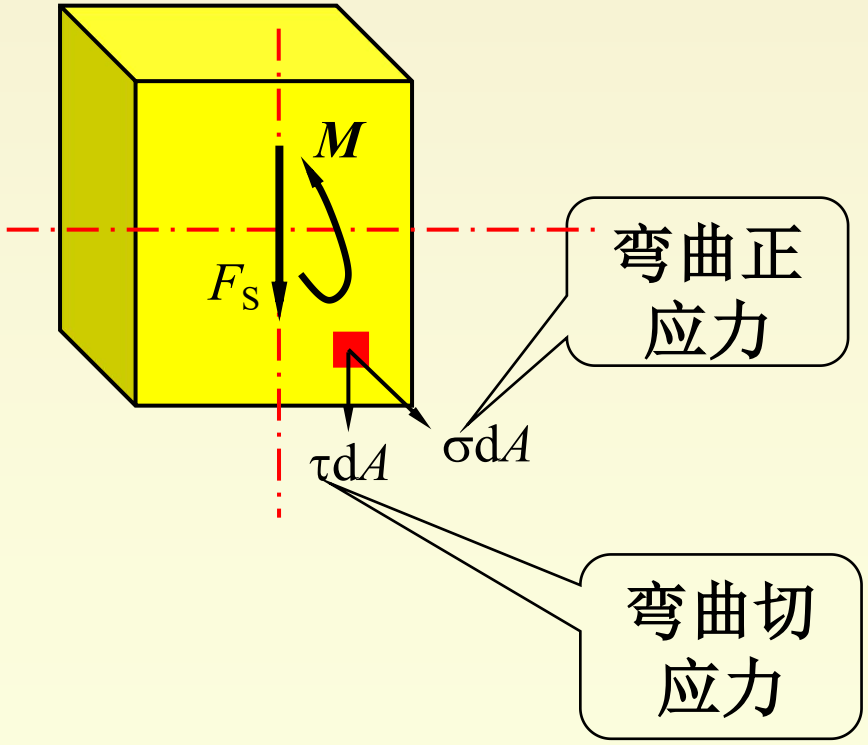
02 纯弯曲

03 纯弯曲时的正应力计算

纯弯曲时的正应力

一、弯曲时横截面上的应力





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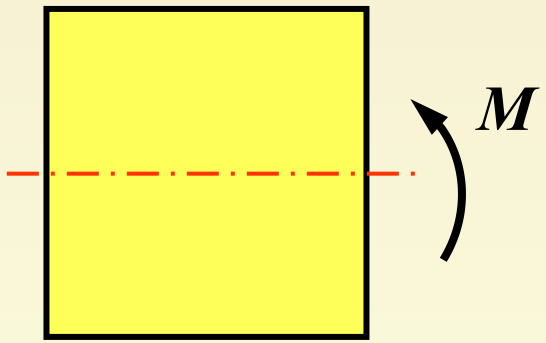
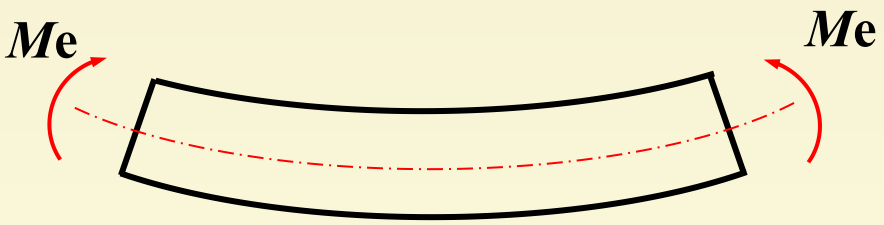
二、纯弯曲

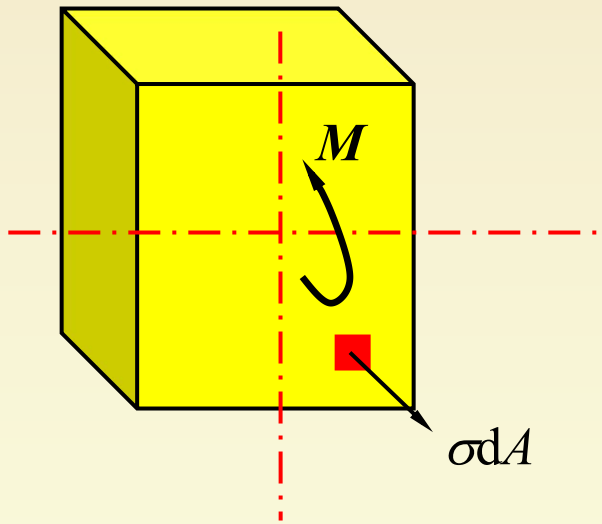




若梁在某段内各横截面的弯矩为常量，剪力为零，则该段梁的弯曲就称为**纯弯曲**。

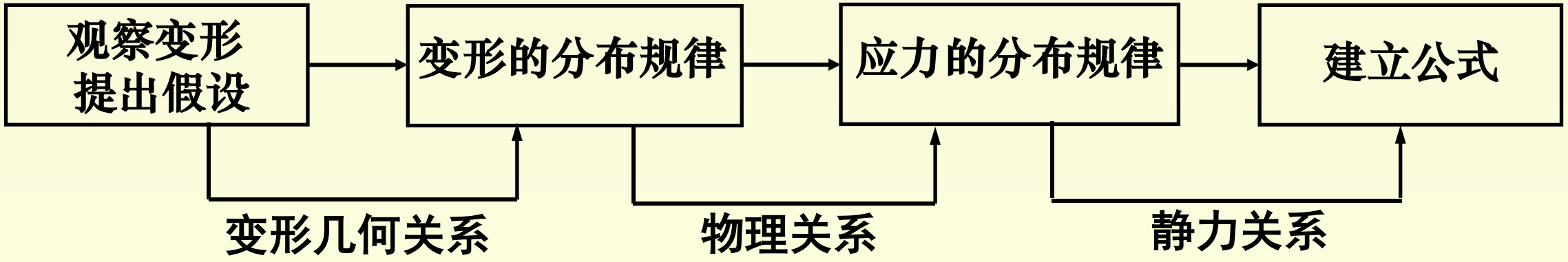
三、纯弯曲时的正应力计算

1、问题分析与研究思路



问题： 横截面的正应力分布情况未知，仅知合成弯矩M。

分析方法：



纯弯曲时的正应力

2、变形几何关系

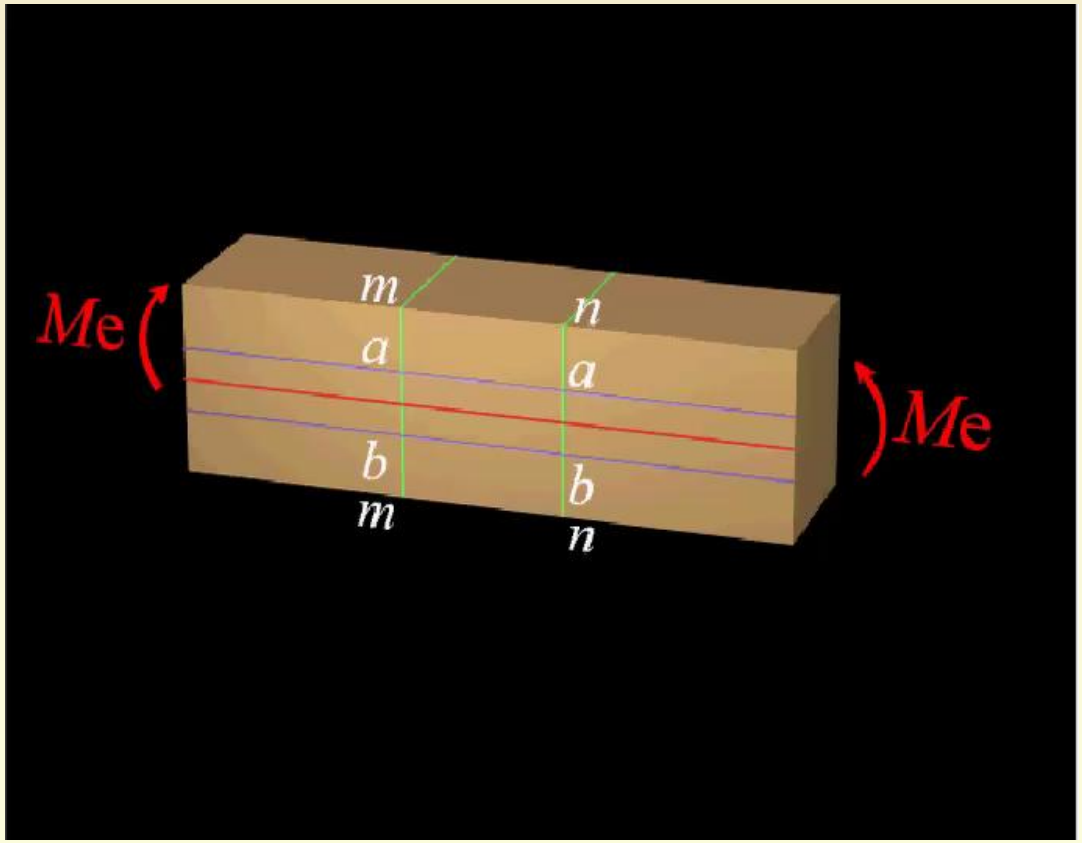
① 变形观察

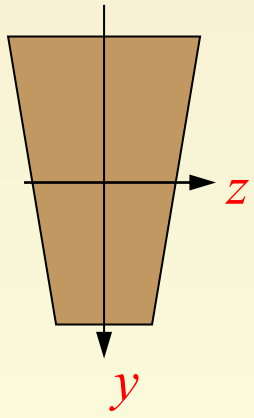
纯弯曲时的正应力

纵向线
- 变为曲线
- 上缩短，下伸长

横向线
- 仍为直线
- 两横线相对转动
- 仍与纵线垂直

横截面
- 上变宽
- 下变窄





纯弯曲时的正应力

② 提出假设

内部变形规律(假设)：

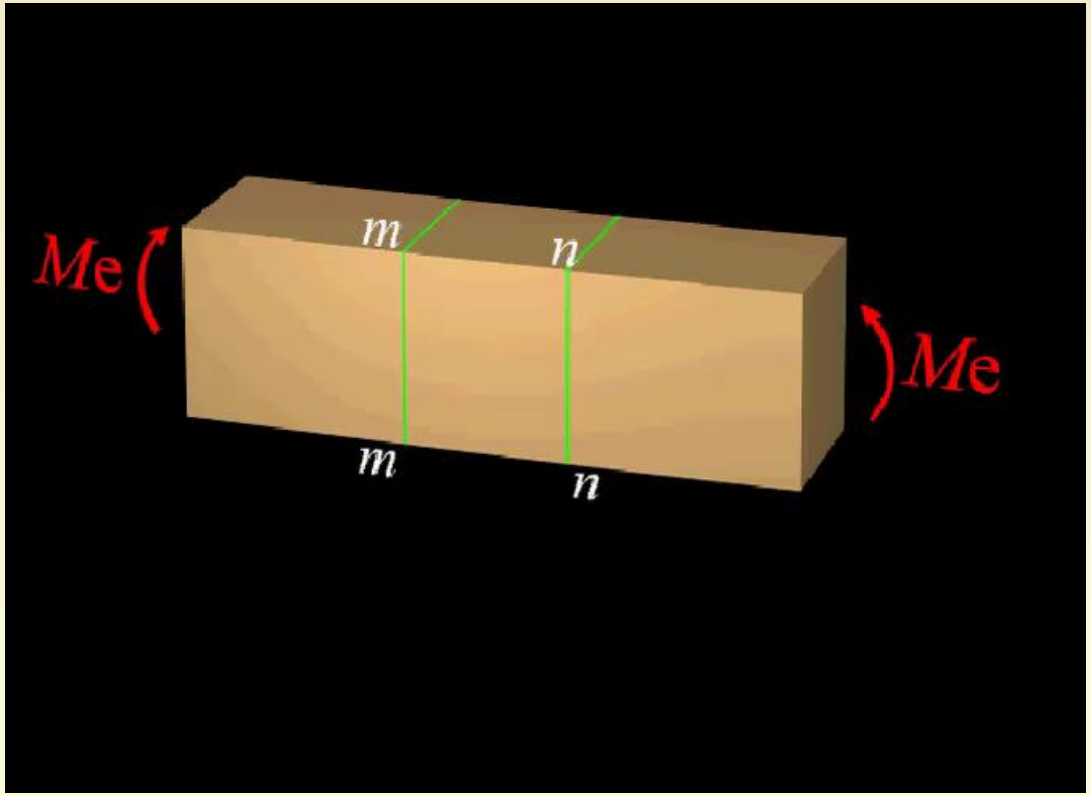
A. 平面假设

变形后横截面仍保持为平面，且

垂直于变形后的梁轴线。

B. 单向受力假设

纵向纤维不相互挤压,只受单向拉压。



纯弯曲时的正应力

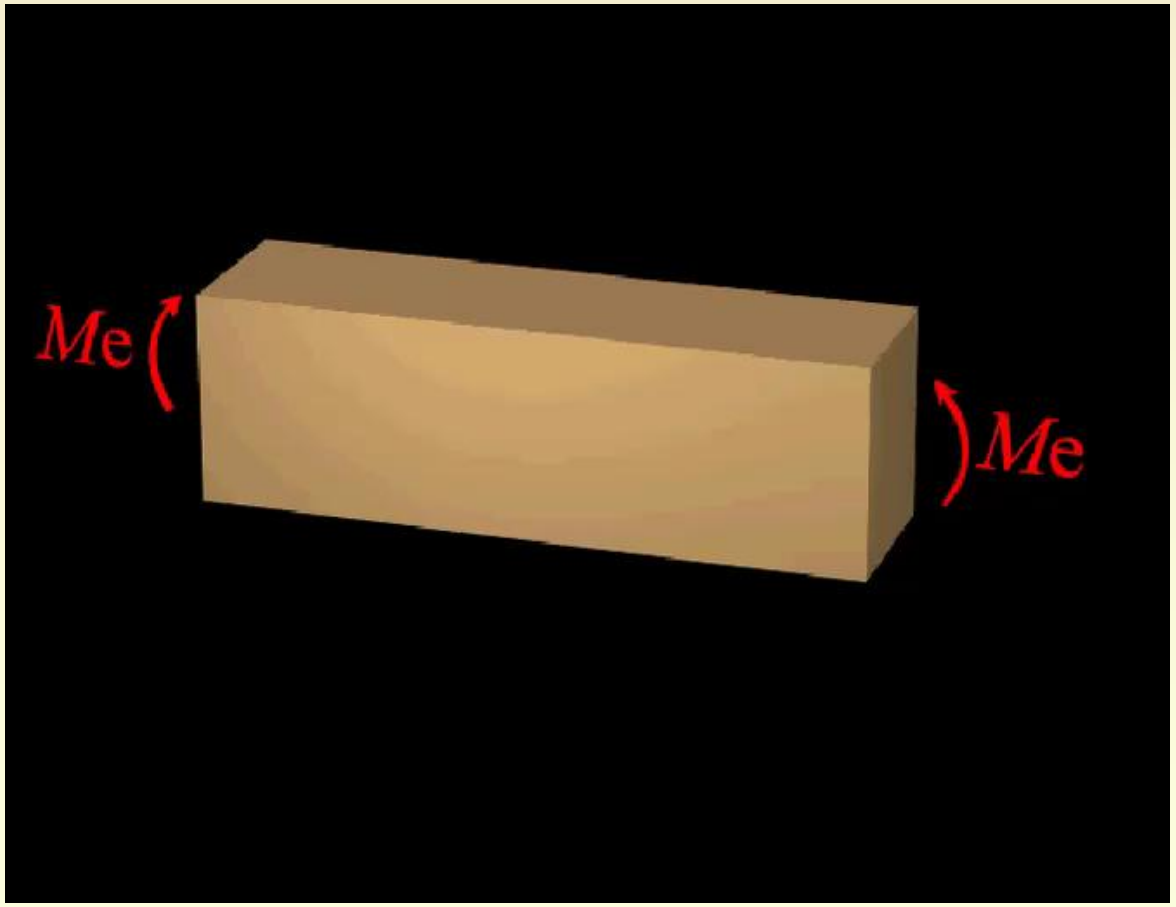
推论：

一侧伸长，一侧缩短

平面假设

存在既不伸长，也不缩短的面

中性层

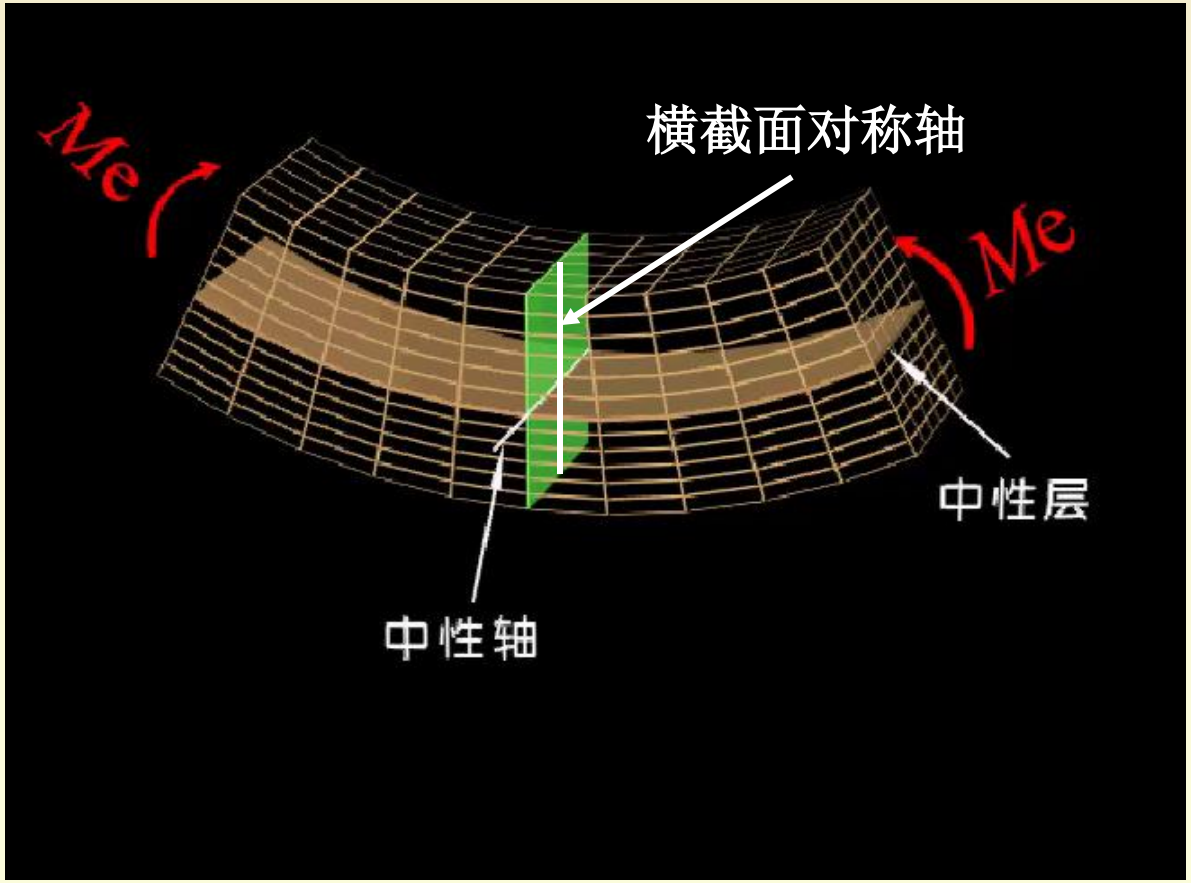


纯弯曲时的正应力

中性轴：

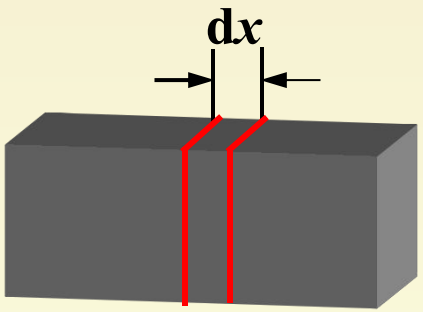
中性层与横截面的交线

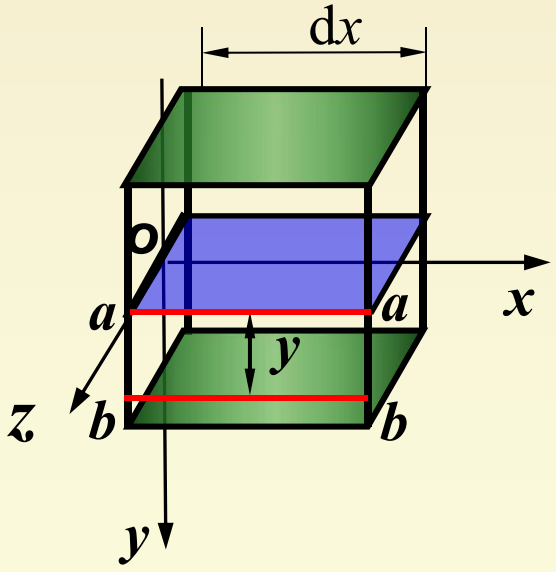
中性轴 ⊥ 横截面对称轴

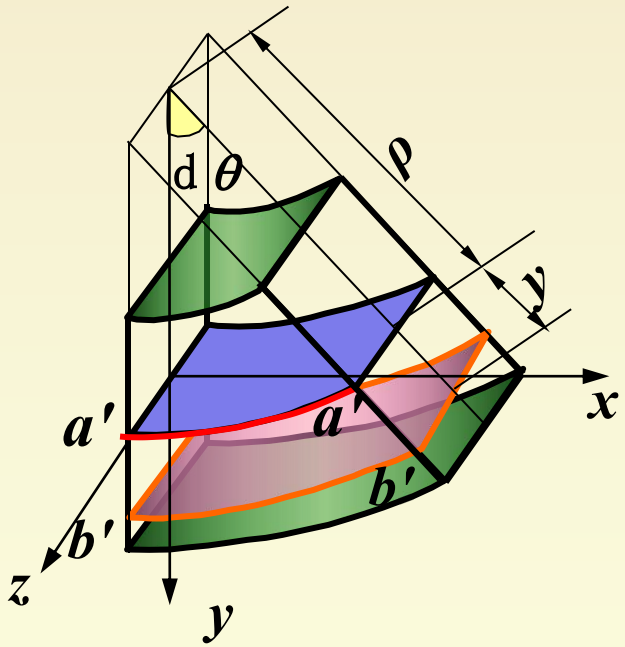


纯弯曲时的正应力

③ 变形分布规律







$$\widehat{b'b'} = (\rho + y)\mathrm{d}\theta$$

$$\overline{bb} = dx = \overline{aa} = \widehat{a'a'} = \rho\mathrm{d}\theta$$

$$\boxed{\varepsilon = \frac{\widehat{b'b'} - \overline{bb}}{\overline{bb}} = \frac{y}{\rho}}$$

纯弯曲时的正应力

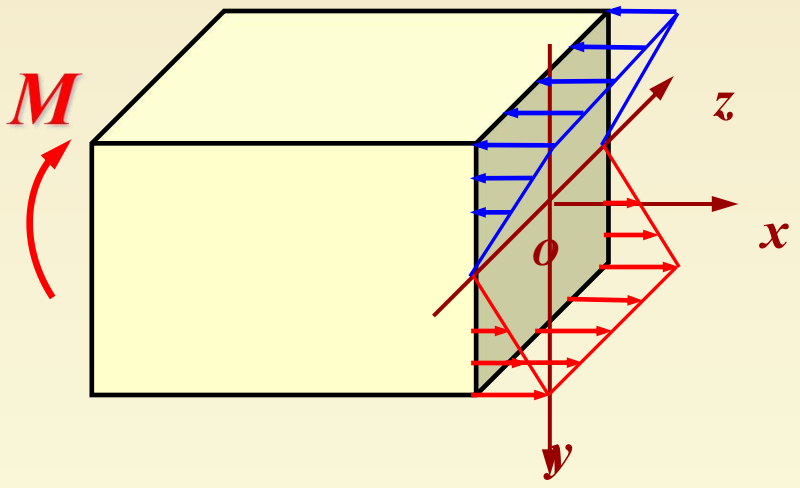
3、物理关系

根据单向受力假设，由轴向拉压胡克定律

$$\sigma = E\varepsilon$$

$$\varepsilon = \frac{y}{\rho}$$

$$\sigma = E\frac{y}{\rho}$$



正应力分布规律：
直梁纯弯曲时横截面上任意一点的正应力，与它到中性轴的距离成正比.

y — 偏离中性轴的坐标值 — 中性轴的位置

ρ — 中性层的曲率半径 — 中性层的曲率半径

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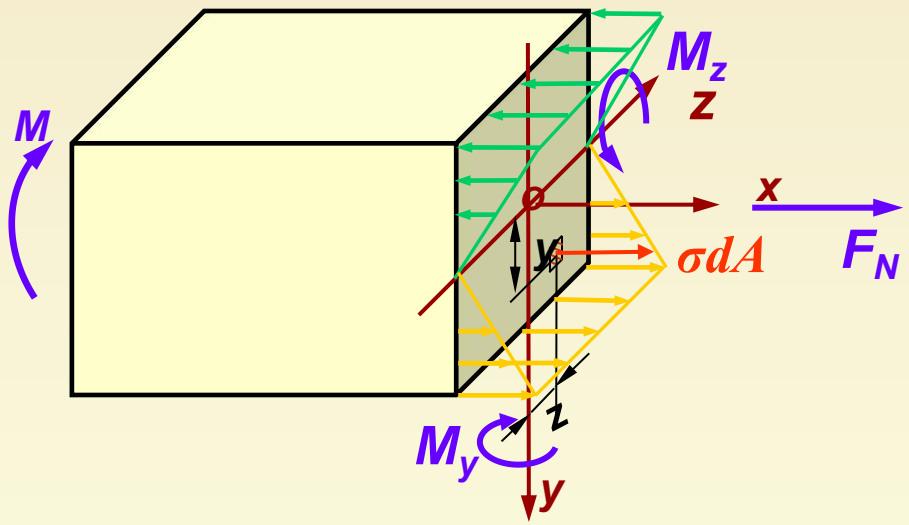
4、静力关系

微内力：σdA

横截面上内力系为垂直于横截面的空间平行力系。



简化后，得到三个内力分量：

$$dF_N = \sigma dA$$

$$dM_y = z\sigma dA$$

$$dM_z = y\sigma dA$$

横截面积分

$$F_N = \int_A dF_N = \int_A \sigma dA = 0$$

$$M_y = \int_A dM_y = \int_A z\sigma dA = 0$$

$$M_z = \int_A dM_z = \int_A y\sigma dA = M$$

纯弯曲时的正应力

将应力式 $\sigma = E\dfrac{y}{\rho}$ 代入 $F_N = \int_A \sigma dA = 0$ ➡ $\int_A E\dfrac{y}{\rho} dA = 0$

➡ $\dfrac{E}{\rho}\int_A y dA = 0$ ➡ $S_z = \int_A y dA = 0$ **中性轴通过横截面形心**

将应力式 $\sigma = E\dfrac{y}{\rho}$ 代入 $M_y = \int_A z\sigma dA = 0$ ➡ $\int_A zE\dfrac{y}{\rho} dA = 0$

➡ $\dfrac{E}{\rho}\int_A yz dA = 0$ ➡ $I_{yz} = \int_A yz dA = 0$ **自然满足**

纯弯曲时的正应力

将应力式 $\sigma = E\dfrac{y}{\rho}$ 代入 $M_z = \int_A y\sigma dA = M$ ➡ $\int_A yE\dfrac{y}{\rho}dA = M$

➡ $\dfrac{E}{\rho}\int_A y^2 dA = M$ ➡ $\dfrac{E}{\rho}I_z = M$ ➡ $\dfrac{1}{\rho} = \dfrac{M}{EI_z}$

⬇

纯弯曲时正应力的计算公式: $\sigma = \dfrac{My}{I_z}$

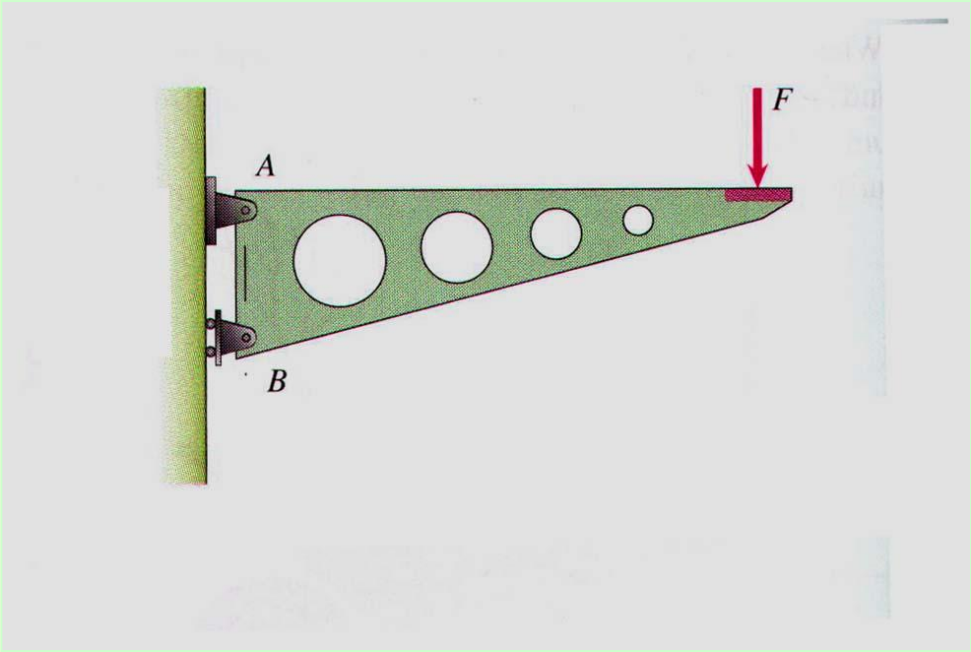
M: 梁横截面上的弯矩;

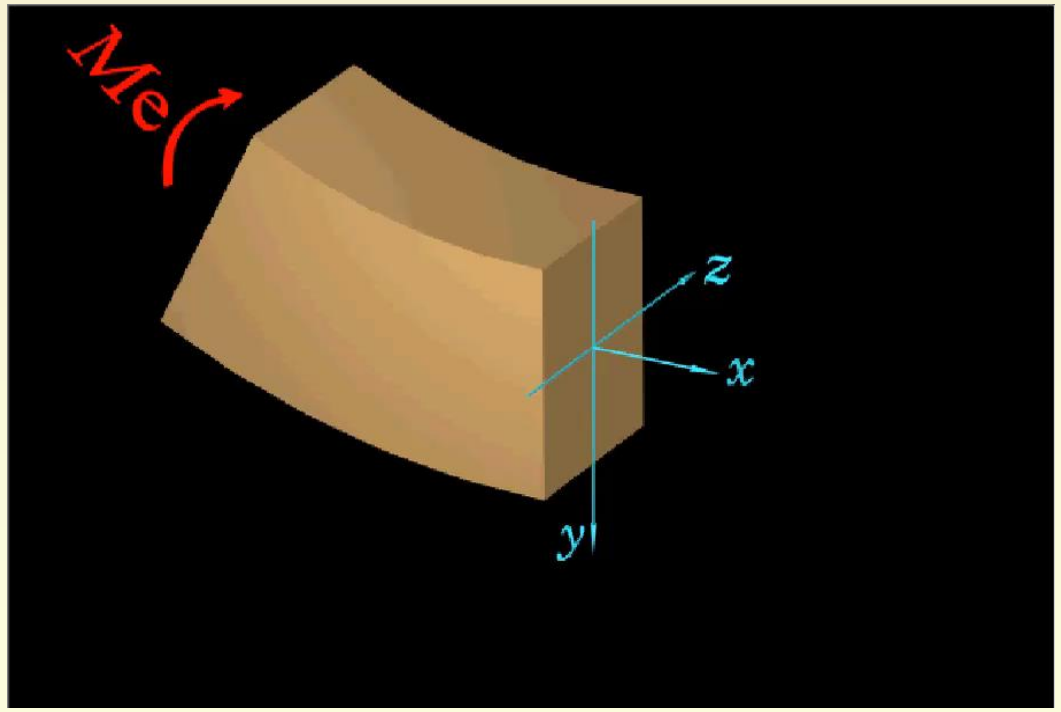
y: 为梁横截面上任意一点到中性轴的距离;

I_z: 为梁横截面对中性轴的惯性矩;

纯弯曲时的正应力

【分析与思考】解释一下托架开孔合理吗？托架会不会破坏？





为降低重量，可在中性轴附近开孔。

谢谢！